

Retail Sentiment Intelligence

Real-Time Brand Health Monitoring via Reddit NLP Pipeline

BITS ZG628T · Dissertation — Post Mid-Semester Progress Presentation

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Dissertation work carried out at **Walmart Global Tech, Bengaluru**

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Agenda

Part A · Mid-Semester Recap (*already presented*)

1. System Architecture & End-to-End Pipeline
2. Vision / Image Processing — Challenge & Mitigation
3. Trust Score & Confidence Calculations
4. ModernBERT — Domain Fine-Tuning Journey
5. Dashboard — Data Population & Sections
6. Notification System — Group-Based Routing

Part B · Post Mid-Semester Work (*new — focus of this presentation*)

7. Review & Validate — Human-in-the-Loop
8. Post Explorer & Filtering
9. Post Lifecycle — Kanban Workflow

Part A · Mid-Semester Recap

Quick walkthrough of what was built and demoed at mid-sem

Recap 1 — System Architecture & Pipeline

From Reddit data to actionable brand intelligence (*mid-sem*)

High-Level System Architecture

Layered view — external sources → pipeline core → AI runtime → storage → serving → clients

```
flowchart TB
    %% --- L1: Sources ---
    subgraph L1["L1 - Data Sources"]
        AS["Arctic Shift API<br/>free · no auth"]
        PR["PRAW / Reddit API<br/>optional OAuth"]
    end

    %% --- L2: Ingestion & Pipeline ---
    subgraph L2["L2 - Ingestion & Processing Pipeline (Python · asyncio)"]
        ING["Ingest<br/>25 subreddits"]
        PRE["Preprocess<br/>clean · langdetect · dedup"]
        TRU["Trust Score<br/>meta + dedup + LLM"]
        ANA["Analyze<br/>text + vision"]
        AGG["Aggregate<br/>hourly / daily"]
        ALR["Alert Engine<br/>spike + severity"]
        ING --> PRE --> TRU --> ANA --> AGG --> ALR
    end

    %% --- L3: AI Runtime ---
    subgraph L3["L3 - AI / ML Runtime"]
        MB["ModernBERT<br/>sentiment · 3-class"]
        BART["BART-MNLI<br/>zero-shot aspects"]
        GEM["Gemma 3 4B<br/>vision · 01lama"]
        FT5["FLAN-T5<br/>reply generation"]
    end

    %% --- L4: Storage ---
    subgraph L4["L4 - Storage (SQLite dev · Cosmos DB prod)"]
        T1["raw_posts"]
        T2["analyses"]
        T3["aggregates"]
        T4["alerts"]
        T5["feedback"]
        T6["notif_log"]
    end

    %% --- L5: Serving ---
    subgraph L5["L5 - Serving Layer · FastAPI :8001"]
        REST["REST API<br/>/api/*"]
        WS["WebSocket<br/>/ws/alerts"]
    end

    %% --- L6: Clients ---
    subgraph L6["L6 - Clients"]
        UI["React Dashboard :5173<br/>Brand Health · Explorer · Review · Insights"]
        EMAIL["Email DL<br/>SMTP"]
        SLACK["Slack<br/>Webhook"]
    end

    %% --- Edges ---
    L1 --> ING
    ANA -. inference .-> L3
    ING --> T1
    ANA --> T2
    AGG --> T3
    ALR --> T4
    T1 & T2 & T3 & T4 --> REST
    T4 -. push .-> WS
    REST --> UI
    WS --> UI
    ALR --> EMAIL
    ALR --> SLACK
    EMAIL & SLACK --> T6
    UI -. corrections .-> T5
    T5 -. few-shot .-> FT5

classDef l1 fill:#FEF3C7,stroke:#D97706,color:#111,stroke-width:1.5px
```

Component Stack Summary

Layer	Stack	Notes
 Data Sources	Arctic Shift API + PRAW (optional)	Free, no auth needed
 Pipeline	Python 3.13 · asyncio scheduler	6-hour cadence + on-demand
 AI Runtime	HuggingFace (offline) + Ollama + Azure OpenAI (opt-in)	Local-first, modular
 Storage	SQLite (dev) / Azure Cosmos DB (prod)	Pluggable backend
 API	FastAPI + WebSocket	Port 8001
 Frontend	React 18 + TypeScript + Vite + Tailwind	Port 5173
		

6-Stage Pipeline — Detailed Flow

```
flowchart TB
    Sched(["Scheduler tick<br/>every 6 h · or manual Run Now"])

    subgraph S1["Stage 1 · INGEST"]
        I1["Arctic Shift API<br/>cursor-based pagination"]
        I2["25 subreddits<br/>Walmart · Spark · Sam's · Pharmacy · Intl · Competitors"]
        I1 --> I2
    end

    subgraph S2["Stage 2 · PREPROCESS"]
        P1["Clean HTML / Markdown"]
        P2["langdetect → English only"]
        P3["MiniLM-L6-v2 embeddings<br/>cosine > 0.92 = duplicate"]
        P1 --> P2 --> P3
    end

    subgraph S3["Stage 3 · TRUST SCORE"]
        T1["Metadata heuristics<br/>age · karma · length · engagement"]
        T2["Dedup / originality score"]
        T3["LLM credibility check<br/>only if 0.3 &lt; meta &lt; 0.8"]
        T4["Weighted<br/>0.4·meta + 0.3·dedup + 0.3·llm"]
        T1 --> T4
        T2 --> T4
        T3 --> T4
    end

    subgraph S4["Stage 4 · ANALYZE"]
        A1["Has image?"]
        A2["Vision · Gemma 3 4B<br/>multi-pass captioning"]
        A3["ModernBERT<br/>sentiment + confidence"]
        A4["BART-MNLI<br/>aspect tags"]
        A1 -- yes --> A2 --> A3
        A1 -- no --> A3
        A3 --> A4
    end

    subgraph S5["Stage 5 · AGGREGATE"]
        G1["Hourly rollups"]
        G2["Daily rollups"]
        G3["Per-subreddit · per-aspect"]
        G1 --> G3
        G2 --> G3
    end

    subgraph S6["Stage 6 · ALERT"]
        L1["Spike detection<br/>volume > 20"]
        L2["Sentiment crash<br/>drop > 0.3 in 6 h"]
        L3["P1 / P2 classification"]
        L4["Notify configured groups"]
        L1 --> L3
        L2 --> L3
        L3 --> L4
    end

    UI(["React Dashboard<br/>live alerts via WebSocket"])

    Sched --> S1 --> S2 --> S3 --> S4 --> S5 --> S6
    S6 -. push .-> UI

    classDef stage fill:#DBEAFE,stroke:#1E40AF,color:#111,stroke-width:1.5px
    classDef sched fill:#FEF3C7,stroke:#D97706,color:#111,stroke-width:1.5px
    classDef ui fill:#DCFCE7,stroke:#166534,color:#111,stroke-width:1.5px
    class S1,S2,S3,S4,S5,S6 stage
    class Sched sched
    class UI ui
```

Data Flow — From Raw Post to Dashboard KPI

How a single Reddit post transforms into a scored, stored, and surfaced artefact

```
flowchart LR
    RP["Raw Post<br/>title · body · media · author · ts"]
```

```
    subgraph CLEAN["Cleaning"]
        C1["Strip HTML / md"]
        C2["Language filter"]
        C3["Dedup vs 24 h window"]
    end
```

```
    subgraph SCORE["Scoring"]
        S1["Trust score<br/>0.0 - 1.0"]
        S2["Sentiment<br/>neg / neu / pos + conf"]
        S3["Aspects<br/>price · service · quality · ..."]
        S4["Vision caption<br/>if image present"]
    end
```

```
    subgraph PERSIST["Persist"]
        D1[("raw_posts")]
        D2[("analyses")]
        D3[("aggregates")]
        D4[("alerts")]
    end
```

```
    subgraph SURFACE["Surface"]
        K1["Brand Health KPIs"]
        K2["Aspect Drilldown"]
        K3["Alert Feed"]
        K4["Post Explorer"]
    end
```

```
    RP --> C1 --> C2 --> C3 --> D1
    D1 --> S1
    D1 --> S2
    D1 --> S3
    D1 --> S4
    S1 --> D2
    S2 --> D2
    S3 --> D2
    S4 --> D2
    D2 --> D3
    D2 --> D4
    D3 --> K1
    D3 --> K2
    D4 --> K3
    D2 --> K4
```

```
classDef raw fill:#FEF3C7,stroke:#D97706,color:#111
classDef clean fill:#E0F2FE,stroke:#0369A1,color:#111
classDef score fill:#E0E9FE,stroke:#5B21B6,color:#111
classDef persist fill:#FEE2E2,stroke:#B91C1C,color:#111
classDef surface fill:#DCFCE7,stroke:#166534,color:#111
```

```
class RP raw
```


Request Sequence — End-to-End

```
sequenceDiagram
    autonumber
    participant Sched as Scheduler
    participant Pipe as pipeline.py
    participant Arc as Arctic Shift
    participant DB as SQLite / Cosmos
    participant LLM as LLM Runtime
    participant Alert as Alert Engine
    participant Notif as Dispatcher
    participant UI as Dashboard

    Sched->>Pipe: tick (every 6 h)
    Pipe->>Arc: GET posts after cursor
    Arc-->>Pipe: JSON batch (n posts)
    Pipe->>Pipe: clean · langdetect · dedup
    Pipe->>DB: INSERT raw_posts

    Note over Pipe,LLM: Trust scoring
    Pipe->>Pipe: metadata + dedup score
    opt uncertain (0.3 < meta < 0.8)
        Pipe->>LLM: credibility prompt
        LLM-->>Pipe: score 0.0 – 1.0
    end

    Note over Pipe,LLM: Analysis
    Pipe->>LLM: ModernBERT (sentiment)
    LLM-->>Pipe: {label, confidence}
    Pipe->>LLM: BART-MNLI (aspects)
    LLM-->>Pipe: aspect tags[]
    alt has image
        Pipe->>LLM: Gemma 3 4B multi-pass
        LLM-->>Pipe: caption text
    end

    Pipe->>DB: INSERT analyses
    Pipe->>DB: UPSERT aggregates

    Pipe->>Alert: run spike + severity check
    Alert->>DB: INSERT alerts
    Alert->>Notif: P1 / P2 → route to groups
    Notif-->>UI: WebSocket push (live)
    UI->>DB: GET /api/brand-health (poll on events)
```

Storage Schema — SQLite / Cosmos DB

Table	Partition Key	Key Fields	Purpose
raw_posts	/subreddit	id, title, body, author_hash, created_utc	Ingested data (privacy-safe)
analyses	/subreddit	post_id, sentiment, confidence, aspects, trust_score	AI analysis results
aggregates	/time_window	subreddit, window, metrics_json	Pre-computed KPIs
feedback	/analyst_id	post_id, correction, reply_text	Human corrections + replies

Recap 2 — Vision / Image Processing

Eliminating hallucination in multimodal retail analysis

The Problem: Image-Only Posts

- **3.9%** of Reddit posts have empty bodies — the complaint lives inside the **image**
- Screenshots of error messages, damaged products, receipts, app glitches
- Text-only pipeline misses these entirely → incomplete brand health

Example

Field	Value
Title	<i>"Can anyone help me? I need this fixed"</i>
Body	<i>(empty)</i>
Image	Screenshot of Walmart app error message

- **Without vision AI** → post scored on title alone (useless)
- **With vision AI** → we extract the error text and understand the complaint

Vision Model Selection — Why Gemma 3 4B

Model	DocVQA	Size	Ollama	Verdict
gemma3:4b (Google)	83	3.3 GB	✓ Yes	✓ SELECTED
LLaVA-1.5 7B	28	4.7 GB	✓ Yes	✗ 3× worse OCR
LLaVA-1.6 8B	75	5.5 GB	✓ Yes	✗ Larger, slower
BLIP-2	N/A	990 MB	✗ No	✗ Caption only
PaliGemma 2 3B	81	~6 GB	✗ No	✗ No Ollama runtime

Why Gemma 3 4B wins:

- Best DocVQA score under 4 GB (reads receipts, screenshots, app screens)
- Google-maintained (USA-based → Walmart policy compliant)
- Reuses existing Ollama infrastructure (`localhost:11434`)

Initial Testing — 75% Failure Rate

Metric	Result	Impact
Overall failure rate	75% (6/8)	Most images get wrong captions
Hallucination rate	50% (4/8)	Model invents false details
Critical hallucinations	37.5%	Fake receipts, fake prices
Correct text extraction	25% (2/8)	Only memes/photos pass

What hallucination looked like

- "Walmart receipt: \$39.99, handwritten 'Damaged Box'" → actually a product page, no price, no handwriting
- "Order status PENDING" → no PENDING text anywhere in image
- "12-pack of 12 fl oz Zero Sugar Dr Pepper" → single 42.3 fl oz regular bottle

Root Cause Analysis

Root Cause	Technical Explanation
768 px fixed resize	Small text becomes unreadable at reduced resolution
4 B parameter limit	Struggles with multi-element reasoning (button + error + context)
No dynamic resolution	Entire image processed at one fixed scale
Hallucination under uncertainty	Invents plausible details rather than admitting it can't see
No context awareness	Confuses screenshots with physical displays

Academic Research — 5 Papers Reviewed (2023–2025)

Paper	Key Technique	Deployable at Walmart?
UReader (Tencent)	Shape-adaptive cropping	✗ China-origin
TextMonkey (USTC)	Shifted window attention	✗ China-origin
DocOwl 1.5 (Alibaba)	Structure-aware parsing	✗ China-origin
InternVL2 (Shanghai AI Lab)	Tile-based processing	✗ China-origin
Qwen2.5-VL (Alibaba)	Dynamic resolution + multimodal RoPE	✗ China-origin

Every single model that solves our problem is from a Chinese lab — blocked by Walmart's vendor policy.

Our Strategy — Take the Ideas, Not the Models

PAPERS SAY:

Use models with native tiling + dynamic resolution

WALMART SAYS:

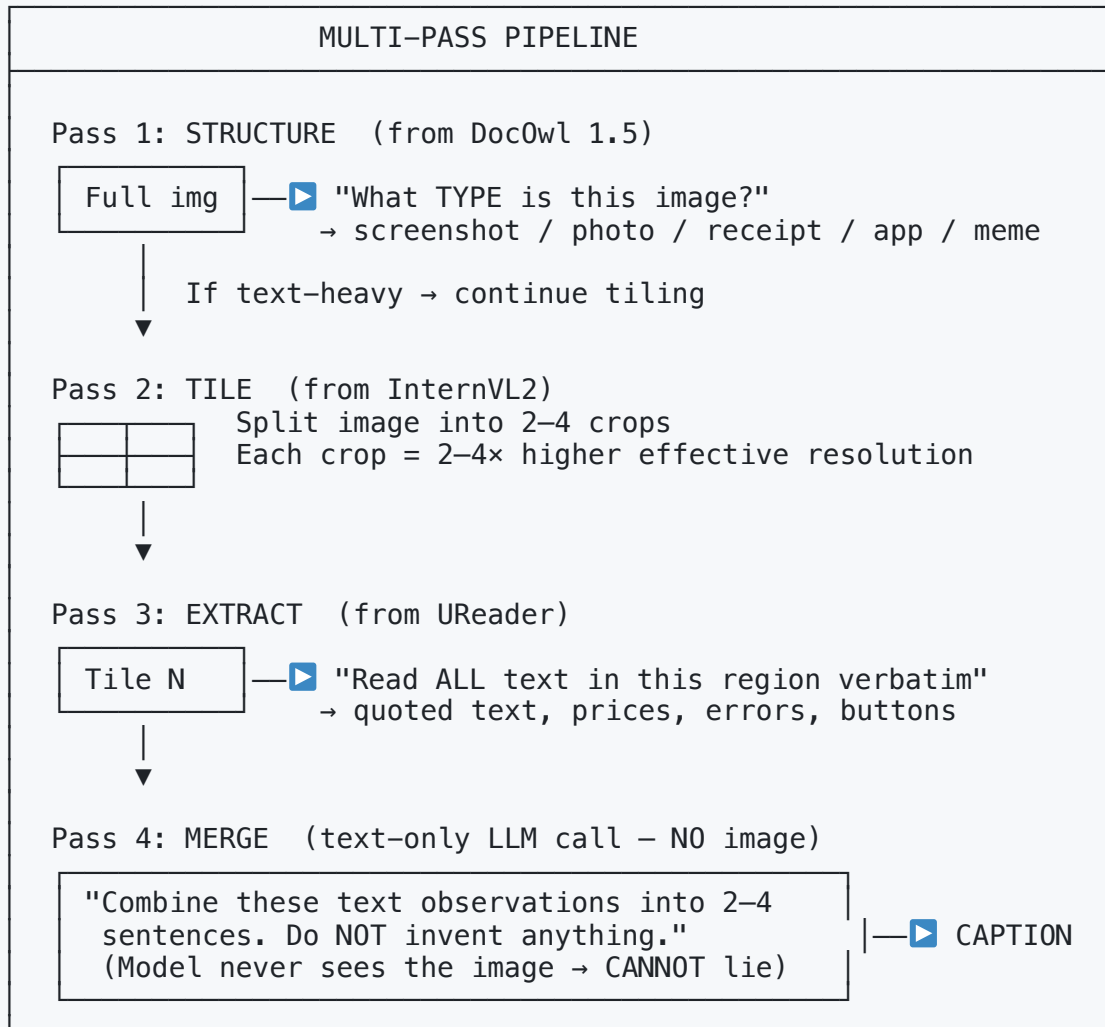
You can't use any of those models (China-origin)

OUR SOLUTION:

Implement their TECHNIQUES as CODE on our compliant model

Technique from Papers	We Implement As
Tile-based processing (InternVL2)	Split image into 2–4 crops ourselves, send each to gemma3:4b
Structure-first parsing (DocOwl 1.5)	First pass asks "what type of image is this?" before reading text
Focused text extraction (UReader)	Dedicated "read ALL text verbatim" prompt per tile
Resolution preservation (Owen2.5-VL)	Tiling gives each crop 2–4× effective resolution without resizing

Multi-Pass Captioning Pipeline (The Fix)



Results — Hallucination Eliminated

Initial 8-image validation

Metric	Before (Single-Pass)	After (Multi-Pass)	Change
Hallucination rate	50% (4/8)	0% (0/8)	↓ 100%
Overall failure rate	75% (6/8)	25% (2/8)	↓ 67%
Correct text extraction	25% (2/8)	75% (6/8)	3× better
Fabricated claims	8 total	0	Eliminated
Avg latency / image	~5 s	~15 s	3× (acceptable)

Scaled validation — 25 images

Verdict	Count	Percentage
 PASS (correct, no hallucination)	22 / 25	88%

Recap 3 — Trust Score & Confidence

| How we validate post credibility and model certainty

Trust Score — Weighted Combination Formula

$$\text{trust_score} = 0.4 \times \text{metadata} + 0.3 \times \text{dedup} + 0.3 \times \text{llm_credibility}$$

Each component is normalized to **[0, 1]** and the final score is clamped.

Component 1: Metadata Heuristics (weight: 0.4)

$$\text{meta} = w_{\text{base}} + w_{\text{age}} \cdot \text{age} + w_{\text{karma}} \cdot \text{karma} + w_{\text{length}} \cdot \text{length} + w_{\text{eng}} \cdot \text{engagement}$$

Signal	Formula	Default Weight
Base floor	constant	0.15
Account age	<code>min(account_age_days / 365, 1.0)</code>	0.20
Karma	<code>min(total_karma / 5000, 1.0)</code>	0.20
Content length	<code>min((len(title) + len(body)) / 200, 1.0)</code>	0.30
Engagement	<code>min(max(reddit_score, 0) / 20, 1.0)</code>	0.15

Credibility Signals — Negative & Positive

Signal Type	Indicator	Score Impact
✗ Negative	Promotional language (≥ 2 phrases)	-0.25
✗ Negative	URL stuffing (≥ 3 links, short text)	-0.20
✗ Negative	Karma/age mismatch (new account, high karma)	-0.20
✗ Negative	Excessive CAPS ($> 40\%$ letters)	-0.15
✗ Negative	New account + promotional	-0.20
✓ Positive	Retail insider terms (≥ 2 : OGP, ASM, CAP2, Spark...)	+0.25
✓ Positive	Long-form organic text (> 600 chars, no links)	+0.15
✓ Positive	Single retail insider term	+0.10

Confidence Score — Model Certainty

Confidence = softmax probability of the predicted class.

How it's calculated

1. ModernBERT outputs **logits** for [negative, neutral, positive]
2. **Softmax** converts to probabilities: $P(\text{neg})$, $P(\text{neu})$, $P(\text{pos})$
3. $\text{confidence} = \max(P(\text{neg}), P(\text{neu}), P(\text{pos}))$

Thresholds used in the system

Threshold	Value	Source
Analysis confidence	≥ 0.7	config/models.yaml
Notification P1	trust ≥ 0.70 AND confidence ≥ 0.80	dispatcher.py
Notification P2	trust ≥ 0.50 AND confidence ≥ 0.60	dispatcher.py

Recap 4 — ModernBERT Fine-Tuning

Domain-specialized long-context sentiment classification

Why ModernBERT over RoBERTa?

Thesis Claim

Long-context, domain-fine-tuned encoders **beat** short-context Twitter-trained baselines on Reddit-flavored retail complaints.

RoBERTa Limitations

- Only **512 token** context → truncates long Reddit posts
- Trained on Twitter → different register from Reddit
- No domain knowledge of retail / Walmart terminology

ModernBERT Advantages

- **8192 token** context (16× longer)
- Supports full Reddit complaint posts without truncation

3-Stage Curriculum Training

Stage	Dataset	Epochs	Purpose
1. Generic Sentiment	TweetEval-sentiment (45 K tweets)	2	Polarity grounding
2. Reddit Register	GoEmotions-3class (54 K Reddit comments)	2	Reddit language patterns
3. Domain Specialization	Walmart-200 (5-fold CV)	up to 15 (patience 3)	Retail fine-tuning

Training Configuration

- `max_length = 1024` tokens (key lever for long-context advantage)
- Effective batch size **32** (per-device BS=8 × grad accum=4)
- Class weights: `neg=0.52`, `neu=1.03`, `pos=8.33` (inverse frequency)

Key Challenges & Mitigations

Challenge	Impact	Mitigation
Synthetic benchmark (77-char bodies)	Cannot test long-context	Built real 200-post benchmark (min 300 chars)
Corp network blocked HuggingFace	Cannot download models	Restart + hotspot + offline triad env vars
Stage 3 macro F1 = 0.40 (catastrophic)	Positive class collapsed	Class weights + oversampling + curriculum
Eval showed F1 = 1.0 (memorization)	Leakage — useless result	Switched to out-of-fold CV predictions
Long-context not showing up	<code>max_length=512</code> defeats purpose	Retrained at <code>max_length=1024</code>

Final Results — ModernBERT vs RoBERTa

(5-fold out-of-fold cross-validated)

Metric	RoBERTa	ModernBERT v2	Improvement
Macro F1 (overall)	0.6272	0.7642	+0.137 (+22%)
F1 negative	0.7967	0.8779	+0.081
F1 neutral	0.6087	0.7480	+0.139
F1 positive	0.4762	0.6667	+0.190
Long-post F1 (≥ 512 tokens)	0.2778	1.0000	+0.722 ✨
Short-post F1 (n=193)	0.6360	0.7619	+0.126
Latency (ms/post, MPS)	6.5 ms	11.9 ms	+5.4 ms

The long-context hypothesis is **proven**: on posts ≥ 512 tokens, ModernBERT scores a

Training Evidence — Artifacts on Disk

Proof that ModernBERT was actually fine-tuned in-house (not a downloaded checkpoint)

Training pipeline

Stage	Script / File	Outp
Data collection	<code>scripts/fetch_real_benchmark.py</code>	<code>data/benchmark_real_200</code> Reddit posts, body 300–360
Human labeling	<code>scripts/label_benchmark.py</code> (interactive)	<code>human_sentiment</code> field per · pos=8)
Curriculum trainer	<code>scripts/train_modernbert_sentiment.py</code>	3-stage checkpoints under <code>models/modernbert_walma</code>
Honest		

Recap 5 — Dashboard & Data Population

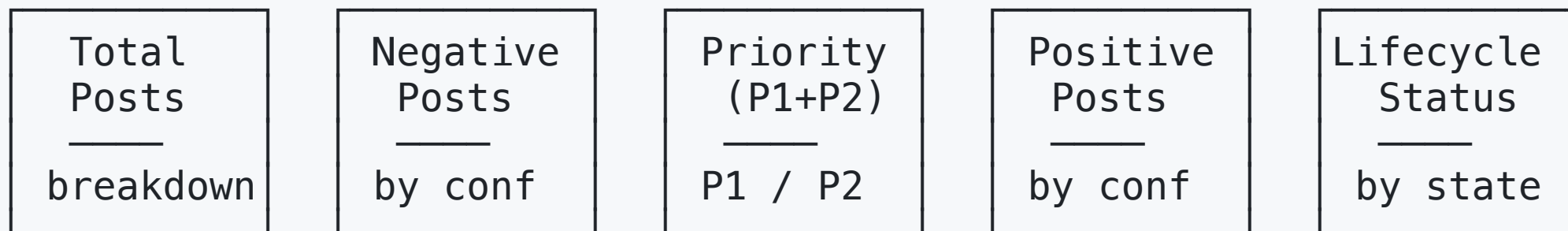
From pipeline output to actionable insights on screen

Dashboard Pages — Navigation Structure

Page	Priority	Purpose	Key Metric
Brand Health	P0	At-a-glance KPIs & trends	Overall sentiment score
Post Explorer	P1	Search / filter all posts	Volume + sentiment distribution
Review & Validate	P0	Correct labels + draft replies	Accuracy improvement
Post Lifecycle	P0	Kanban workflow (triaged → resolved)	Resolution rate
Insights	P1	AI-generated competitor analysis	Issue rankings
Pipeline Control	P1	Monitor & trigger runs	Jobs, cursors, health

Brand Health — KPI Tiles with Embedded Breakdowns

5 tiles, each with a main value + lifecycle breakdown rows + dual navigation:



Each tile supports

- **Single-click** → navigate to Post Explorer (pre-filtered)
- **Dropdown** → dual nav to Post Explorer OR Review & Reply

Data sources

- `/api/brand-health?range=today` — aggregated sentiment counts

How Data Populates the Dashboard

Pipeline Output → SQLite Tables → FastAPI Endpoints → React Components

SQLite Table	API Endpoint	UI Component
<code>analyses</code> (sentiment, confidence, aspects, trust)	<code>/api/brand-health</code> , <code>/api/posts</code> , <code>/api/segments</code>	KPI Tiles, Trend Charts, Aspect Heatmap
<code>aggregates</code> (hourly/daily)	<code>/api/brand-health?</code> <code>range=week/month</code>	Trend Line, Volume Ticker
<code>alerts</code> (spike, severity)	<code>/api/alerts</code> + WebSocket push	Alert Feed (real-time)
<code>feedback</code> (corrections)	<code>/api/review/*</code> , <code>/api/review/{id}/draft</code>	Review Queue, Draft Replies

Recap 6 — Notification System

| Group-based routing for P1 / P2 priority posts

Notification System — Architecture

```
Pipeline analyzes post → classify_priority(trust, confidence)
```

```
├── Not P1/P2? → Skip (no notification)
```

```
├── P1 or P2? → Find matching notification groups
```

```
│   ├── Group matches subreddit + priority filter?
```

```
│       ├── Email DL configured → Send email
```

```
│       └── Slack channel configured → Send Slack
```

```
│   └── Log to notification_log table (audit trail)
```

Priority Classification

- **P1:** `trust ≥ 0.70` AND `confidence ≥ 0.80` (high-signal, immediate action)
- **P2:** `trust ≥ 0.50` AND `confidence ≥ 0.60` (review-worthy, lower urgency)

Notification Configuration Page (Frontend)

Admin UI at `/notifications` allows:

-  Create notification groups (name, subreddits, email DL, Slack)
-  Quick-add subreddits by category (Walmart core, Spark, Pharmacy, International, Sam's Club, Competitors)
-  Priority filter: choose P1, P2, or both per group
-  Enable / disable groups with toggle
-  **Test (dry-run)** — simulates notification without sending
-  View delivery log — audit trail of all sent notifications
-  Delete groups

API Endpoints (8 total)

Method	Path	Purpose
--------	------	---------

Part B · Post Mid-Semester Work

New capabilities built after mid-sem — analyst workflows, lifecycle tracking, competitor insights

What's new since mid-sem

#	Capability	Why it was added
1	Review & Validate queue	Close the loop on AI mistakes; harvest labels for retraining
2	Post Explorer & filtering	Give analysts fast search across all ingested posts
3	Post Lifecycle (Kanban)	Track a complaint from <i>New</i> → <i>Resolved</i> with SLA visibility
4	Insights & Competitor Analysis	Benchmark Walmart vs Target / Costco / Kroger on shared aspects

Post-Midsem 1 — Review & Validate



Human-in-the-loop correction and reply generation

Review Queue — Human-in-the-Loop

Purpose: Analysts correct AI labels and draft replies to negative posts.

Workflow

1. Queue shows posts sorted by priority (P1 first)
2. Analyst reviews sentiment + aspects (correct if wrong)
3. Click "Generate Drafts" → two reply options generated

Dual-Draft Reply Generation

Draft A
Smart Composer
(keyword extraction
+ phrase pools)

Draft B
FLAN-T5-base
(multi-temp
sampling + scorer)

Post-Midsem 2 — Post Explorer



Search, filter, and deep-dive into analyzed posts

Post Explorer — Features

Purpose: Browse all analyzed posts with powerful filtering.

Filters Available

- **Sentiment:** negative / neutral / positive
- **Confidence:** threshold slider
- **Trust score:** threshold slider
- **Subreddit:** multi-select from 25 tracked
- **Aspect:** delivery, product_quality, returns, customer_support, pricing, app_website
- **Date range:** today, week, month, custom
- **Text search:** full-text across titles & bodies

Per-Post Card Shows

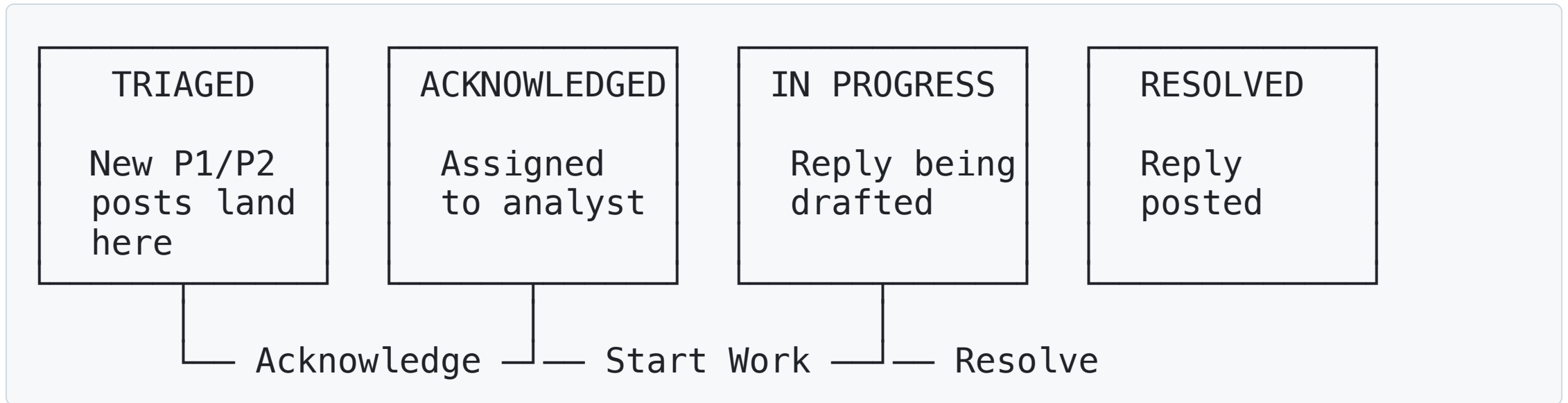
Post-Midsem 3 — Post Lifecycle



Kanban workflow from triage to resolution

Post Lifecycle — Kanban Board

States: Triaged → Acknowledged → In Progress → Resolved



Resolve Modal (2-step flow)

Step 1: Save action note + optional LLM-drafted reply

- "Save reply & open Reddit" — copies reply to clipboard, opens thread
- OR "Resolve (no reply needed)" — closes without posting

Post-Midsem 4 — Insights & Competitor Analysis



AI-generated strategic intelligence from raw data

Insights Page — What It Provides

1. Issue Rankings

- Top negative issues ranked by **volume × severity × recency**
- Grouped by aspect (delivery, pricing, app, etc.)
- Trend arrows showing improvement or deterioration

2. Competitor Pulse

- Compare sentiment: Walmart vs **Costco, Target, Amazon**
- Cross-mentioned posts (e.g., "Walmart vs Costco" comparisons)
- Subreddit-level breakdown per competitor

3. LLM Summarization

- Natural-language summaries of the week's top themes

Post-Midsem 5 — Results & Future Work



What we achieved and what's next

Key Achievements — Post Midsem

Area	Achievement	Evidence
ModernBERT	Macro F1: 0.6272 → 0.7642 (+22%)	5-fold OOF CV
Long-context	Long-post F1: 0.28 → 1.00 (+722%)	7 posts ≥ 512 tokens
Vision	Hallucination: 50% → 0%	8 + 25 image validation
Vision	Text extraction: 25% → 75%	Multi-pass pipeline
Pipeline	25 subreddits, every 6 h (automated)	Arctic Shift API
Dashboard	7 pages, real-time WebSocket alerts	React + FastAPI
Notifications	Group-based P1/P2 routing	Email + Slack channels
Lifecycle	Full Kanban with 2-step resolve	Triage → Resolved

Technical Stack Summary

Layer	Technology	Key Decision
Backend	Python 3.13 + FastAPI + SQLite	Free, local-first, modular
Frontend	React 18 + TypeScript + Vite + Tailwind	Modern SPA, responsive
Sentiment	ModernBERT (fine-tuned, 1024 tokens)	Domain-specialized, offline
Aspects	BART-MNLI (zero-shot classification)	No training needed
Vision	Gemma 3 4B via Ollama (multi-pass)	Policy compliant, no hallucination
Reply Gen	FLAN-T5 + Smart Composer (dual draft)	Learning loop via feedback

Future Work

- ☐ Azure Cosmos DB migration (production-grade storage)
- ☐ Twitter / X integration as second data source
- ☐ 3-seed ensemble for ModernBERT (+0.01–0.03 F1, tighter variance)
- ☐ Gemma 3 12B upgrade for remaining edge-case images
- ☐ Spanish language support (bilingual retail communities)
- ☐ Azure AD authentication for multi-user access
- ☐ Automated retraining pipeline (feedback loop → model updates)
- ☐ Slack bot integration for inline notification responses
- ☐ Blind 25-post recheck for ModernBERT defensibility

Thank You

Retail Sentiment Intelligence

Post Mid-Semester Presentation

Questions?

Appendix — Reference Documentation

Topic	Reference File
System Architecture	ARCHITECTURE.md
Pipeline & Tools	PIPELINE_AND_TOOLS.md
Dashboard Design	DASHBOARD_DESIGN.md
Requirements (frozen)	REQUIREMENTS.md
Implementation Plan	IMPLEMENTATION_PLAN.md
ModernBERT Journey	docs/MODERNBERT_JOURNEY.md
Vision Pipeline Story	docs/LIVE_DEMO_VISION_PIPELINE.md
Vision Model Comparison	docs/VISION_MODEL_COMPARISON_v1.md
Model Comparison (Thesis)	docs/MODEL_COMPARISON.md